United States Patent	12407213
Kind Code	B2
Date of Patent	September 02, 2025
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Rotating device

Abstract

A rotating device according to the present invention comprises a motor, a connection terminal electrically connecting the motor with an external device, and a housing accommodating the motor and the connection terminal. The connection terminal includes a bent portion bent in a direction crossing an insertion direction or a removal direction of the connection terminal. The housing includes a contact surface in contact with the bent portion.

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Appl. No.: **18/742181**

Filed: **June 13, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20240333087 A1	Oct. 03, 2024

Foreign Application Priority Data

JP	2018-102696	May. 29, 2018
JP	2018-102697	May. 29, 2018
JP	2018-102698	May. 29, 2018

Related U.S. Application Data

division parent-doc US 17057943 US 12051962 WO PCT/JP2019/005534 20190215 child-doc US 18742181

Publication Classification

Int. Cl.: **H02K5/22** (20060101); **H01R13/04** (20060101); **H02K7/116** (20060101); **H02K11/21** (20160101)

U.S. Cl.:

CPC **H02K5/225** (20130101); **H01R13/04** (20130101); **H02K7/1166** (20130101); **H02K11/21** (20160101); H02K2211/03 (20130101)

Field of Classification Search

CPC: H02K (11/21); H02K (7/11); H02K (2211/03)

USPC: 310/71

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional Application of U.S. application Ser. No. 17/057,943 filed on Nov. 23, 2020, which was a national stage entry of PCT/JP2019/005534, filed on Feb. 15, 2019 and claims benefit of Japanese Application Nos. 2018-102696, 2018-102697 and 2018-102698, all filed on May 29, 2018, of which the entire disclosures of which are hereby incorporated herein by reference.

FIELD
(1) The present invention relates to a rotating device.
BACKGROUND
(2) A rotating device is known including a motor and a housing accommodating the motor, configured to transmit rotational power to an external device, and including a crank-shaped terminal connecting the motor with the external device (see Patent Literature 1, for example).
(3) Another rotating device is known including a motor, a board provided with wires, and a flexible board electrically connecting the motor with the board provided with wires, and the flexible board includes lands at a terminal portion connected by soldering (see Patent Literature 2, for example).
(4) Still another rotating device is known including a gear rotated by a motor to transmit rotational power to an external device, a sensor configured to detect the number of rotations or a rotational angle of the gear, and a housing accommodating these components. The sensor includes a brush configured to be rotated together with the gear and a wiring board provided with a wire pattern in electrical contact with the brush (see Patent Literature 3, for example).
CITATION LIST
Patent Literature
(5) Patent Literature 1: Japanese Laid-open Patent Publication No. 2011-041382 Patent Literature 2: Japanese Laid-open Patent Publication No. 2015-216844 Patent Literature 3: Japanese Utility Model Application Laid-open No. H01-123451
SUMMARY
Technical Problem
(6) The terminal of the rotating device disclosed in Patent Literature 1 above may not be reliable enough to withstand an external force exerted in a removal direction. The structure of the rotating device disclosed in Patent Literature 2 above may not be reliable enough to keep the flexible board firmly connected. The structure of the rotating device disclosed in Patent Literature 3 has a room for improvement in the disposition of the wiring board to prevent degradation of the sensor performance.
(7) The present invention treats any one of the above-described problem as an example of problems, and an object of the present invention to provide a rotating device capable of achieving

higher reliability or higher performance.
Solution to Problem
(8) In order to achieve the above object, the present invention is grasped by the following constitution. A rotating device comprises a motor, a connection terminal electrically connecting the motor with an external device, and a housing accommodating the motor and the connection terminal. The connection terminal includes a bent portion bent in a direction crossing an insertion direction or a removal direction of the connection terminal. The housing includes a contact surface in contact with the bent portion.
(9) A rotating device according to the present invention comprises a motor, a board provided with a wire, and a flexible board electrically connecting the board provided with the wire with the motor. The flexible board and the board provided with the wire are overlapped via a bonding member. The flexible board includes a first land including a hole. The bonding member is in contact with a second land included in the board provided with the wire through the hole.
(10) A rotating device according to the present invention comprises a gear, a motor configured to rotate the gear, a sensor configured to detect number of rotations or a rotational angle of the gear, and a housing accommodating the gear, the motor, and the sensor. The sensor includes a brush configured to be rotated together with the gear, and a board provided with an arc-shaped wire in electrical contact with the brush. The board provided with the wire is provided with a first hole disposed inside of the arc-shaped wire and rotatable with respect to a rotating shaft of the gear, and a second hole allowing passage of a protruding portion provided to the housing.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a perspective view of a rotating device according to an embodiment with a first casing and a second casing being separated.
(2) FIG. **2** is an exploded perspective view of the rotating device according to the embodiment.
(3) FIG. **3**A is a plan view of the rotating device according to the embodiment with the second casing removed from the rotating device.
(4) FIG. **3**B is a bottom view of the second casing.
(5) FIG. **4**A is a sectional view illustrating the disposition of a board provided with wires.
(6) FIG. **4**B is a sectional view illustrating a first support.
(7) FIG. **4**C is a sectional view illustrating a second support.
(8) FIG. **4**D is a sectional view illustrating a third support.
(9) FIG. **5** is a side view of a terminal.
(10) FIG. **6**A is a cross-sectional view taken along line A-A in FIG. **5**.
(11) FIG. **6**B is a cross-sectional view illustrating a connection terminal according to a modification.
(12) FIG. **7**A is a perspective view illustrating a fixing part in the first casing for fixing connection terminals.
(13) FIG. **7**B is a perspective view illustrating the fixing part in the second casing for fixing the connection terminals.
(14) FIG. **7**C is a schematic view illustrating the fixing part for fixing the connection terminal.
(15) FIG. **8** is a perspective view illustrating a holding member configured to hold the connection terminals.
(16) FIG. **9** is a perspective view illustrating a bonding state between the board provided with wires and a flexible board.
(17) FIG. **10**A is a sectional view illustrating the bonding state between the board provided with wires and the flexible board.

(18) FIG. **10**B is a diagram illustrating a portion of a first planar portion of the flexible board.
(19) FIG. **11**A is a plan view of the rotating device according to another embodiment with a second casing removed from the rotating device.
(20) FIG. **11**B is a perspective view of a flexible board included in the rotating device according to another embodiment.
(21) FIG. **11**C is a plan view of a rotating device according to still another embodiment with the second casing being removed.
(22) FIG. **12** is a perspective view illustrating a connection terminal provided at the connector portion of the rotating device according to another embodiment.

DESCRIPTION OF EMBODIMENTS

(23) The following describes modes (hereinafter, called "embodiments") for carrying out the present invention will be described in detail based on the accompanying drawings. The same components are denoted by the same reference signs throughout the description of the embodiments.
(24) FIG. **1** is a perspective view of a rotating device according to an embodiment with a first casing and a second casing being separated. FIG. **2** is an exploded perspective view of the rotating device according to the embodiment. FIG. **3**A is a plan view of the rotating device according to the embodiment with the second casing being removed. FIG. **3**B is a bottom view of the second casing. FIG. **4**A is a sectional view illustrating the disposition of a board provided with wires. FIG. **4**B is a sectional view illustrating a first support. FIG. **4**C is a sectional view illustrating a second support. FIG. **4**D is a sectional view illustrating a third support.
(25) This rotating device **1** according to the embodiment is used in, for example, a vehicle air conditioning system (not illustrated), and can control a rotational operation of a louver for controlling, for example, an air volume. As illustrated in FIG. **1**, the rotating device **1** includes a housing **2** accommodating a functional part including, for example, a motor **3**, an output gear **5**, a plurality of transmission gears **6**, and a sensor **7**. These components will be described later.
(26) The housing **2** includes separable first and second casings **21** and **22**. Each of the first casing **21** and the second casing **22** is made of a resin material, such as polypropylene, polyethylene terephthalate, or acrylonitrile butadiene styrene (ABS).
(27) The first casing **21** includes a first surface portion **210** serving as a bottom surface of the housing **2** (FIG. **4**A), a first sidewall **211** provided at an outer circumferential portion of the first surface portion **210**, and an opening **212**. As illustrated in FIGS. **2** and **3**A, the first casing **21** includes a protruding portion **218** at an inner surface of a first surface portion **210** for use in positioning a board **700** provided with wires included in the sensor (angle sensor) **7**.
(28) The second casing **22** includes a second surface portion **220** serving as a top surface of the casing **2** (FIG. **4**A), a second sidewall **221** provided at an outer circumferential portion of the second surface portion **220**, and an opening **222** surrounded by the second sidewall **221**. As illustrated in FIG. **3**B, the second casing **22** includes a first step portion **201** on an inner surface of the second surface portion **220** configured to support the board **700** provided with wires. As illustrated in FIG. **4**D, the first casing **21** includes a protruding portion **201***a* configured to hold the board **700** provided with wires together with the first step portion **201**.
(29) As illustrated in FIGS. **3**B, **4**A, and **4**B, the second casing **22** includes a tubular support portion **228** configured to rotatably support an end portion of a rotating shaft **51** of the output gear **5** and having a tip end portion serving as a second step portion **202** configured to support the board **700** provided with wires. As illustrated in FIG. **3**B, an outer circumference of the second casing **22** is provided with four mounting portions **23**, **24**, **25**, and **26** for mounting the rotating device **1** in a predetermined position when the rotating device **1** is installed in, for example, the air conditioning system.
(30) In this way, the housing **2** is configured by connecting the first casing **21** including the opening **212** to the second casing **22** including the opening **222** in a state where the openings **212** and **222**

oppose each other.
(31) A plurality of projections (hereinafter referred to as engagement projections **213**) are integrally formed at the first sidewall **211** of the first casing **21**, and the projections correspond to a plurality of engagement portions **223** of the second casing **22**. The engagement portions **223** are integrally formed with an outer circumference of the second sidewall **221** of the second casing **22** such that the engagement portions **223** extend toward the first casing **21**. The engagement projections **213** are engaged with the engagement portions **223**. Some of the engagement portions **223** include a hole (hereinafter referred to as an engagement step portion (or step portion) **224**). Some of the engagement projections **213** are engaged with such engagement step portions (step portions) **224**.
(32) The first casing **21** and the second casing **22** are put together and the engagement projections **213** are engaged with the engagement portions **223**. Accordingly, the first casing **21** and the second casing **22** are firmly integrated to configure the housing **2** capable of accommodating the functional part including various components.
(33) In the present embodiment, the first casing **21** is provided with the engagement projections **213**, and the second casing **22** is provided with the engagement portions **223**. However, the first casing **21** may be provided with the engagement portions **223**, and the second casing **22** is provided with the engagement projections **213**.
(34) As illustrated in FIGS. **1** and **2**, a part of the first side wall **211** of the first casing **21** and a part of the second side wall **221** of the second casing **22** are respectively provided with projecting portions **219** and **229** corresponding to each other.
(35) In the present embodiment, the projecting portions **219** and **229** project in a direction of a rotating shaft of the motor **3** extending. The projecting portions **219** and **229** are joined to form a connector portion **20** (see FIG. **4**A), a plurality of connection terminals **4** to be described later being arranged inside the connector portion. The connector portion **20** includes a fixing part **215** (see FIG. **7**C) capable of firmly holding the connection terminals **4**. The connector portion **20** is connected with an external connector via the connection terminals **4** held by the fixing part **215**, and the motor **3** to be described later can be electrically connected with an external device. Details of the fixing part **215** will be described later.
(36) As illustrated in FIGS. **1** to **3**A, the rotating device **1** according to the present embodiment includes, as various components configuring the functional part, at least the motor **3**, the output gear **5** configured to mechanically output the rotation of the motor **3** to an external device, the transmission gears **6** configured to transmit the rotation of the motor **3** to the output gear **5**, and the sensor **7** configured to detect the rotational angle or the number of rotations of the output gear **5**. The rotating device **1** can control the rotation of the motor **3** based on the rotational angle or the number of rotations of the output gear **5** detected by the sensor **7**.
(37) The plurality of transmission gears **6** include a first transmission gear **61** and a second transmission gear **62** each configured to include multiple stages. The rotation of the motor **3** is transmitted from a worm gear **32** mounted on a rotating shaft **31** (FIG. **3**A) and engaged with the first transmission gear **61**, to the output gear **5**, via the second transmission gear **62**. Accordingly, the rotating shaft **51** of the output gear **5** can rotate a driven member coupled with, for example, a louver (not illustrated) of an air conditioning system.
(38) The following describes specific mounting structures of the components included in the functional part with reference to FIGS. **5** to **10** in addition to FIGS. **1** to **4**D. FIG. **5** is a side view of the connection terminal **4**. FIG. **6**A is a cross-sectional view taken along line A-A in FIG. **5**. FIG. **6**B is a cross-sectional view illustrating a connection terminal according to a modification. FIG. **7**A is a perspective view illustrating the fixing part **215** in the first casing **21** for fixing the connection terminals **4**. FIG. **7**B is a perspective view illustrating the fixing part **215** in the second casing **22** for fixing the connection terminals **4**. FIG. **7**C is a schematic view illustrating the fixing part **215** for fixing the connection terminals **4**. FIG. **8** is a perspective view illustrating a holding member configured to hold the connection terminals **4**. FIG. **9** is a perspective view illustrating a bonding

state between the board provided with wires and the flexible board. FIG. **10**A is a sectional view illustrating the bonding state between the board provided with wires and the flexible board. FIG. **10**B is a diagram illustrating a portion of a first planar portion of the flexible board.
(39) (Motor **3**)
(40) As illustrated in FIGS. **1** to **3**A, the motor **3** is a driving source for rotating the output gear **5**. In the present embodiment, a direct-current (DC) motor is used as the motor **3**. As illustrated in FIG. **2**, the motor **3** includes a body **30** including an outer shell (frame) formed in a quadrangular prismatic outer shape including curved corners, the rotating shaft (shaft) **31**, the worm gear **32** mounted on the rotating shaft **31**, and a pair of motor terminals **33**, **33**.
(41) (Transmission Gears **6**)
(42) The transmission gears **6** are gears configured to transmit the rotation of the rotating shaft **31** of the motor **3** to the output gear **5** at a predetermined reduction ratio (gear ratio). In the present embodiment, as described above, the transmission gears **6** include the first transmission gear **61** and the second transmission gear **62** both include multiple stages.
(43) More specifically, as illustrated in FIG. **2**, the transmission gears **6** in the embodiment include the first transmission gear **61** including a first large diameter portion **611** and a first small diameter portion **612**, and the second transmission gear **62** including a second small diameter portion **621** and a second large diameter portion **622**. The first small diameter portion **612** moving together with the first large diameter portion **611** meshes with the second large diameter portion **621** of the second transmission gear **62**, and the second small diameter portion **622** moving together with the second large diameter portion **621** meshes with the output gear **5**.
(44) Since the first transmission gear **61** and the second transmission gear **62** are interposed between the motor **3** and the output gear **5**, the rotation of the rotating shaft **31** of the motor **3** can be transmitted to the rotating shaft **51** of the output gear **5** at a predetermined gear ratio. The transmission gears may include the worm gear **32** mounted on the rotating shaft **31** of the motor **3**, in addition to the first transmission gear **61** and the second transmission gear **62**.
(45) In the present embodiment, the transmission gears **6** including the first and second transmission gears **61** and **62** are used to transmit the rotation of the rotating shaft **31** of the motor **3** to the output gear **5** at an appropriate reduction ratio using a small space in the housing **2**. the first and second transmission gears **61** and **62** includes the multi-stage. However, the structure of the transmission gears **6** is not limited to the structure described in the present embodiment, and may be modified as appropriate to include, for example, a plurality of single-stage transmission gears.
(46) (Output Gear **5**)
(47) As illustrated in FIG. **4**A, the output gear **5** includes the rotating shaft **51** serving as an output shaft and a disc-shaped gear body **52** integrally formed with the rotating shaft **51**. The rotating shaft **51** includes an end portion **512** rotatably supported by the second surface portion **220** of the second casing **22** and an output part **510** disposed opposite to the end portion **512** and including tooth row **511** formed on the inner surface. The output part **510** is disposed in the first casing **21**.
(48) The gear body **52** is located at an outer side of the output part **510**. The gear body **52** includes teeth **513** formed on the outer circumferential surface and meshing with the second small diameter portion **622** of the second transmission gear **62**. The rotation of the motor **3** is thus transmitted to the output gear **5**.
(49) For example, a drive shaft provided with splines can be fitted to the output part **510**. As the drive shaft is rotated, for example, a louver of an air-conditioning system for using a vehicle can be rotated.
(50) As illustrated in FIG. **4**A, the rotating shaft **51** of the output gear **5** includes a step portion **515** configured to support, between the step portion **515** and the second casing **22**, the board **700** provided with wires, the sensor **7** to be described later having the wires. In other words, the board **700** provided with wires of the sensor **7** is provided with a first hole **710** (see FIGS. **2** and **9**) allowing the rotating shaft **51** of the output gear **5** to pass through the first hole **710**. When the

rotating shaft **51** is inserted in the first hole **710**, the board **700** provided with wires is supported by the step portion **515**. Accordingly, the first hole **710** is fixed to the rotating shaft **51**.
(51) The rotating device **1** is not necessarily directly coupled to a drive shaft to be rotated with the output gear **5**. For example, another gear such as a reduction gear may be interposed between the rotating device **1** and the drive shaft to be rotated.
(52) (Sensor **7**)
(53) As described above, for example, to control the rotational operation of the louver of a vehicle air-conditioning system in a predetermined state, the rotating device **1** needs to detect, for example, the rotational angle of the output gear **5** by using the sensor **7**.
(54) As illustrated in FIG. **2**, the sensor **7** includes the board **700** provided with wires and a brush **75**. The brush **75** is made of an electrically conductive material and is mounted on a surface of the output gear **5** in the second casing **22** side to be rotated with the output gear **5**. As illustrated in FIG. **9**, the board **700** provided with wires includes an arc-shaped first wire (hereinafter referred to as a wire pattern) **701** and an arc-shaped second wire (hereinafter referred to as a wire pattern) **702** in electrical contact with contacts **751** and **752** of the brush **75**, respectively.
(55) The board **700** provided with wires is made of, for example, an epoxy resin including a thickness of approximately 300 μm to approximately 1600 μm, and includes a rectangular-shaped portion at one side and a circular-shaped portion at the other side. As illustrated in FIG. **9**, first wire pattern **701** having a high-resistance and second wire pattern **702** having a low resistance are formed in an arc shape about the first hole **710**. In other words, the first wire pattern **701** and the second wire pattern **702** are arranged concentrically with respect to the first hole **710**.
(56) Opposite end portions of the outer first wire pattern **701** undergo electrically pattern connection with land portions **703***a* and **703***c* (second lands) disposed in a edge side of the area having a rectangular-shape. In the same manner, a portion of the second wire pattern **702** disposed inside the first wire pattern **701** undergoes electrically pattern connection with a land portion **703***b* (second land) disposed between the land portions **703***a* and **703***c*. As the detail will be described later, the land portions **703** are soldered to first lands **803**, which is the land portions formed at the flexible board **8** serving as a board. The land portions **703***a*, **703***b*, and **703***c* may be collectively referred to as second lands **703**.
(57) The first wire pattern **701** and the second wire pattern **702** configure a rotary variable resistor. In other words, as contact positions of the contacts **751** and **752** of the brush **75** change in the circumferential direction in accordance with the rotation of the output gear **5**, the resistance value of a path from the one land portion **703***a* at one end to the land portion **703***c* at the other end changes.
(58) When, for example, the contact positions of the brush **75** change in the circumferential direction while a certain voltage is applied between the land portion **703***a* at one end and the land portion **703***c* at the other end, the voltage between the land portion **703***a* and the land portion **703***c* varies. The relation between the change in the contact positions of the brush **75** and the change in the voltage is linearly proportional and thus the rotational angle of the output gear **5** can be detected from the change in voltage.
(59) As illustrated in FIG. **3**A, the board **700** provided with wires is provided with, in addition to the first hole **710**, a second hole **720** and a third hole **730** disposed at positions of the vertexes of an isosceles triangle with the first hole **710** being the apex (see FIGS. **2** and **9**).
(60) The second hole **720** allows passage of the protruding portion **218** (FIG. **2**) provided to the first casing **21**. The second hole **720** includes an elongated circular shape to provide a certain interval from the protruding portion **218** including a circular column shape.
(61) The third hole **730** is formed for use in positioning the flexible board **8** in the soldering process to be described later.
(62) As described above, the second casing **22** includes the first step portion **201** and the second step portion **202** configured to support the board **700** provided with wires. In the present

embodiment, the first step portion **201** is disposed close to the third hole **730** in the second casing **22**.
(63) Accordingly, the board **700** provided with wires is supported at three points, at a first support a**1**, a second support a**2**, and a third support a**3**. In other words, the first support a**1** is configured by the step portion **515** provided to the rotating shaft **51** of the output gear **5** and the second step portion **202** provided to the tubular support portion **228** of the second casing **22**. The second support a**2** is configured by the protruding portion **218** provided to the first casing **21** and a tubular portion **203** provided to the second casing **22**. The third support a**3** is configured by the first step portion **201** provided to the second casing **22** and the protruding portion **201***a* provided to the first casing **21**. The board **700** provided with wires is supported at these three points and thus can be disposed in the housing **2** more accurately parallel to the housing **2**.
(64) At the first support a**1**, a interval is provided between the rotating shaft **51** of the output gear **5** and the first hole **710** in the radial direction for a interval fit. At the second support a**2**, a interval is provided between the protruding portion **218** and the second hold **720** in the radial direction for a interval fit. At the third support a**3**, a interval is provided between the board **700** provided with wires and the first step portion **201** and the protruding portion **201***a* in the direction of the rotating shaft **51** for a interval fit.
(65) As described above, the first hole **710** disposed at the center of the arc-shaped first and second wire patterns **701** and **702** provides a interval fit for the rotating shaft **51** of the output gear **5**, and the protruding portion **218** provided at the housing **2** passes through the second hole **720** at the board **700** provided with wires.
(66) Specifically, the intervals provided between the rotating shaft **51** of the output gear **5** and the first hole **710** and between the protruding portion **218** and the second hole **720** fall within certain tolerances. In addition, the intervals provided between the board **700** provided with wires and each of the protruding portion **201***a*, the first step portion **201**, and the second step portion **202** fall within certain tolerances.
(67) Accordingly, this structure can reduce the difference between the center of the two wire patterns **701** and **702** configuring a variable resistor and the center of rotation of the brush **75** as much as possible, and the board **700** provided with wires can be accurately positioned and fixed in the housing **2**. Accordingly, the functions of the sensor **7** can be fully implemented without reducing the linearity (linear proportional relation) between the change in contact positions of the brush **75** relative to the wire patterns **701** and **702** and the change in voltage.
(68) The linearity (linear proportional relation) is determined by the rotational angle of the output gear **5** and the resistance value of the variable resistor. Specifically, the linearity is determined by the rotational angle of the brush **75** and the resistance value of the wire patterns **701** and **702**. Aligning the first hole **710** of the board **700** provided with wires with the rotating shaft **51** of the output gear **5** can increase the linearity.
(69) (Connection Terminals **4**)
(70) As illustrated in FIG. **1**, the connection terminals **4** are terminals connected to an external connector connected to the rotating device **1**, and the connection terminals **4** are disposed in the connector portion **20**. As illustrated in FIGS. **2**, **3**A, and **9**, the rotating device **1** according to the present embodiment includes five connection terminals **4**. Two terminals are connected to the motor **3** and three terminals are connected to the board **700** provided with wires of the sensor **7** via the flexible board **8** to be described later.
(71) As illustrated in FIG. **5**, the connection terminals **4** include a crank shape including a first extending portion **41** (a portion of the connection terminal **4**), a bent portion **40**, and a second extending portion **42** (another portion of the connection terminal **4**).
(72) The bent portion **40** is bent in a direction crossing the insertion direction or the removal direction of the connector portion **20** to or from an external connector. The bent portion **40** includes a first corner **401** and a second corner **402** bent in opposite directions of a direction orthogonal to

the removal direction, and a middle portion **403** connecting the first corner **401** with the second corner **402**.
(73) In other words, the connection terminals **4** disposed in the connector portion **20** are formed in a crank shape including the bent portion **40**, and including the first extending portion **41** and the second extending portion **42** extending in the removal direction. The first extending portion **41** extends in the removal direction with an end extending toward outside of the housing **2** from the first corner **401**. The second extending portion **42** is formed parallel to the first extending portion **41** and extends from the second corner **402** toward the inside of the housing **2**. An end portion of the first extending portion **41** is electrically connected with an external device, and an end of the second extending portion **42** is electrically connected with the motor **3** or the sensor **7**.
(74) As illustrated in FIG. **6**A, the connection terminals **4** according to the present embodiment include a rectangular cross-section, and are formed with a bent metal member having a bar shape and flat angles. Accordingly, this structure facilitates forming of the bent portion **40** and can increase the strength of the connection terminals **4**. This structure can also prevent increase in material costs and manufacturing costs of the connection terminals **4** and can in turn reduce the overall cost of the rotating device **1**. The cross-section of the connection terminals **4** is not limited to a rectangular shape, and may be a square shape as illustrated in FIG. **6**B, may be a pentagonal shape or any other polygonal shapes with more sides than those of a pentagon, or may be a circular shape.
(75) The connection terminals **4** according to the present embodiment are disposed such that the bent portion **40** is in contact with contact surfaces **216** formed at the housing **2**. In other words, as illustrated in FIG. **7**A, a wall portion **219***a* is formed at deep inside the projecting portion **219** of the first casing **21** configuring the connector portion **20**. A first fixing part **215***a* including a first contact surface **216***a* is provided at the opening **212** side of the wall portion **219***a*. On the other hand, as illustrated in FIG. **7**B, the second surface portion **220** is positioned at deep inside the projecting portion **229** of the second casing **22** configuring the connector portion **20**. a second fixing part **215***b* including a protruding shape and including a second contact surface **216***b* is provided at the second surface portion **220**. As illustrated in FIG. **7**C, when the first casing **21** and the second casing **22** are put together, the contact surfaces **216** is formed inside the housing **2** and has the contact surfaces **216** by the first fixing part **215***a* and the second fixing part **215***b* opposing each other.
(76) As illustrated in FIG. **7**C, the second casing **22** includes a protruding portion **22***c*, a step portion **22***d*, and a recessed portion **22***e*. The protruding portion **22***c*, the step portion **22***d*, and the recessed portion **22***e* forms an engagement portion **22***f*. The first fixing part **215***a* of the first casing **21** is formed with a protruding portion includes a protruding shape and configures an engagement portion **21***f* (see FIG. **1**) to be engaged with the engagement portion **22***f*. The engagement portion **21***f* of the first casing **21** and the engagement portion **22***f* of the second casing **22** are engaged with the first corner **401** and the second corner **402**, respectively.
(77) As described above, the housing **2** according to the present embodiment includes the contact surfaces **216** in contact with the bent portion **40** of the connection terminals **4**. Accordingly, the force exerted in the insertion direction or the removal direction of the connection terminals **4** can be received by the surfaces of the housing **2**, and the resistance of the connection terminals **4** to an external force exerted in the insertion direction or the removal direction can be easily and reliably increased.
(78) As illustrated in FIG. **7**C, the fixing part **215** includes a recessed portion **217** to be engaged with the bent portion **40** of the connection terminals **4**. In other words, the recessed portion **217** is formed to create a interval between the first fixing part **215***a* provided at the first casing **21** and the second fixing part **215***b* provided at the second casing **22** such that the connection terminals **4** can be arranged at the interval. The bent portion **40** of the connection terminals **4** is engaged with the recessed portion **217** formed in this manner. Accordingly, the bent portion **40** of the connection

terminals **4** is easily engaged with the recessed portion **217** by disposing the connection terminals **4** at the fixing part **215**, and the connection terminals **4** can be easily and securely fixed.
(79) As illustrated in FIGS. **2** and **8** to **10**B, the rotating device **1** according to the present embodiment includes a rectangular plate-shaped holding member **9** configured to hold a plurality of (five) connection terminals **4** in a line. The holding member **9** includes as through-holes **91** as the number of connection terminals **4** to be held. Since number of connection terminals **4** is five in the present embodiment, the holding member **9** includes five through-holes **91** disposed at certain intervals in the longitudinal direction as illustrated in FIG. **8**.
(80) The holding member **9** is disposed in contact with an inner surface of the wall portion **219***a* where the first fixing part **215***a* of the first casing **21** is provided. Accordingly, the five connection terminals **4** can be disposed easily and collectively at the fixing part **215** of the housing **2**, and the resistance of the five connection terminals **4** to an external force exerted in the removal direction can be collectively increased.
(81) (Flexible Board **8**)
(82) The flexible board **8** as a board is made of a flexible film. The flexible board **8** includes a structure including an adhesive layer and a conductor, for example. The adhesive layer is provided on a film (resin substrate) including a thickness of, for example, approximately 12 μm to 50 μm. The conductor includes a thickness of, for example, approximately 12 μm to approximately 50 μm and is printed or pasted onto the adhesive layer. The film is formed of an insulating resin material, such as polyimide or polyester, for example. The conductor is formed of a metal material, such as copper, for example. The adhesive layer is formed of an epoxy resin or an acrylic resin, for example. The flexible board **8** described above can restore a form before being bent even when being bent at an angle of 90 degrees or larger.
(83) Accordingly, if vibrations are generated by the rotation of the motor **3** and the vibrations are transmitted to the flexible board **8**, such vibrations can be absorbed. The flexible board **8**, which is electrically connected with the motor **3**, the sensor **7**, and the connection terminals **4** joined by a bonding member such as solder **78**, can avoid application of high stress to the joint portions between these components, and thus can prevent cracks or damages on the joint portions and can prevent the resulting breakage and disconnection.
(84) As illustrated in FIG. **2**, the flexible board **8** mainly includes three planar portions **81**, **82**, and **83**. In other words, the flexible board **8** includes a first planar portion **81** including the first lands **803** connected with the board **700** provided with wires, a second planar portion **82** connected to the motor terminals **33**, and a third planar portion **83** connecting the first planar portion **81** with the second planar portion **82**. The second planar portion **82** configures a side portion of the flexible board **8** and includes a substantially semi-circular recess **802** to let an end portion of the motor **3** pass through.
(85) The first planar portion **81** faces the board **700** provided with wires. As illustrated in FIG. **10**B, the first planar portion **81** includes a portion connected with the connection terminals **4**, a portion **804** connected with the board **700** provided with wires via the first lands **803**, and holes **805**. The portion **804** connected with the board **700** provided with wires is a surface opposing the board **700** provided with wires. As illustrated in FIG. **10**A, a bent portion **800** is formed between an end portion **81***a* (first end portion) of the portion connected with the connection terminals **4** and an end portion **81***b* (second end portion) of the portion connected with the board **700** provided with wires.
(86) misalignment in the removal direction of the connection terminals **4** in soldering the first planar portion **81** with the board **700** provided with wires via the first lands **803** are absorbed by providing the bent portion **800**. To determine an approximate soldering position of the flexible board **8**, a hole provided at an end portion of the first planar portion **81** may be used for alignment with the third hole **730** provided at the board **700** provided with wires for use in positioning as illustrated in FIG. **9**.
(87) As illustrated in FIGS. **10**A and **10**B, the first lands **803** at the first planar portion **81** of the

flexible board **8** include the holes **805**. When the flexible board **8** is soldered, the solder **78** used as the bonding member is in contact with the second lands **703** at the board **700** provided with wires via the holes **805**.

(88) In soldering the flexible board **8** with the board **700** provided with wires by using the solder **78**, the solder **78** may be deposited from a portion of each first land **803** to a portion of each second land **703**. This configuration can facilitate bonding the flexible board **8** with the board **700** provided with wires by use of the solder **78** and can easily guide the solder **78** to the holes **805** of the first lands **803** to bring the solder **78** into contact with the second lands **703**.

(89) As illustrated in FIG. **9**, a plurality of third wires **705** connected to the first wire pattern **701**, and a wire **704** connected to the second wire pattern **702** are formed at the board **700** provided with wires. As illustrated in FIG. **10**B, a plurality of recesses **806** are provided at a side portion **84** of the flexible board **8** on the extensions of the third wires **705** and the fourth wire **704**.

(90) By providing the holes **805** and the recesses **806** at the flexible board **8**, two types of ridges L**1** and L**2** are formed at the solder **78** straggling the flexible board **8** and the board **700** provided with wires. By increasing lengths of two types of ridges L**1** and L**2**, the adhesion between the flexible board **8** and the board **700** provided with wires can be increased, and it can prevent lifting of the flexible board **8**. The recesses **806** may be used for positioning the third wires **705** and the fourth wire **704** connected to the second wire pattern **702** relative to a plurality of holes **805**.

(91) Although the flexible board **8** according to the present embodiment includes a simple structure, this simple structure implements direct bonding between the first lands **803** of the flexible board **8** and the second lands **703** of the board **700** provided with wires using the solder **78** through the holes **805**. Therefore, adhesion is increased. Accordingly, the flexible board **8** can be easily and firmly bonded to the board **700** provided with wires without lift. Specifically, in the present embodiment, as described above, the flexible board **8** includes in particular the bent portion **800**. The bending of the bent portion **800** applies force to the flexible board **8** in a lifting direction from the board **700** provided with wires bonded with the flexible board **8**, but the sufficient adhesion can prevent lifting of the flexible board **8** and reduce the possibilities of, for example, disconnection.

(92) Described next is a rotating device **1** according to another embodiment. FIG. **11**A is a plan view of the rotating device according to another embodiment with the second casing being removed. FIG. **11**B is a perspective view of a flexible board included in the rotating device according to another embodiment. FIG. **11**C is a plan view of a rotating device according to still another embodiment with the second casing being removed. In FIGS. **11**A to **11**C, components including the same shape or the same functions as those described in the embodiment above are given the same reference signs and, for example, their detailed descriptions may be omitted.

(93) The rotating device **1** illustrated in FIG. **11**A differs from the rotating device **1** described above in that the housing **2** includes a different shape, a flexible board **810** electrically connecting the board **700** provided with wires with the motor **3** includes a different shape, and slits **820** as an example of a slit part are formed at the flexible board **810**.

(94) In other words, as illustrated, the first casing **21** configuring the housing **2** of the rotating device **1** according to another embodiment includes both end portions on the shorter sides of the housing **2**. Tip portions **93** are formed at the both end portions on the shorter sides of the housing **2** and protrude outward a hole (joint hole) **94** is formed at each of tip portions **93**, and a fasteners (not illustrated) are inserted in the holes **94**.

(95) As illustrated in FIG. **11**B, the flexible board **810** included in the rotating device **1** according to another embodiment includes a plurality of lands (hereinafter referred to as “terminal-side lands”) **832** connected with a plurality of connection terminals **4** (see FIG. **11**A) and includes a plurality of first lands **831** to be connected with the board **700** provided with wires. The terminal-side lands **832** are connected with the first lands **831** via wires **833**. The plurality of terminal-side lands **832** are each connected with the connection terminals **4** and also connected with, for example, lands on

a board mounted with a certain electronic component. On the other hand, the plurality of first lands **831** are connected with, for example, terminals of the sensor **7** and terminals of the motor **3**.
(96) As illustrated in FIG. **11**B, at the flexible board **810***y*, the slits **820** are provided from the terminal-side lands **832** toward the first lands **831**. The slits **820** are provided in parallel with the wires **833** in a direction of the rotating shaft **31**. As illustrated, slit **820** includes a certain length and includes holes (circular holes) **820***a* at both end portions to prevent cracks from forming from the end portions. Two end portions **820***b* and **820***c* oppose each other in a direction crossing the slits **820** (e.g., in a direction orthogonal to the extension direction of the wires **833**). slit **820** is disposed between the two end portions **820***b* and **820***c*. Accordingly, the flexible board **810** includes discontinuous surfaces or discontinuous portions including the slits **820** in a direction crossing the slits **820**.
(97) If, for example, vibrations are generated by the rotation of the motor **3** or an external force is exerted, the flexible board **810** can absorb such vibrations or force to some extent. In the present embodiment, by providing the slits **820** at the flexible board **810**, it can prevent damages such as breakage of the wires **833** or can prevent disconnection at the first lands **831** or at the terminal-side lands **832** if, for example, a stress is applied to the flexible board **810**.
(98) Vibrations in the direction of the rotating shaft **31** of the motor **3** (extension direction of the wires **833**) can be reliably absorbed by bending of the flexible board **810** and vibrations in the direction orthogonal to the direction of the rotating shaft **31** of the motor **3** or in the extension direction of the wires **833** can be absorbed by relative displacement of the two end portions **820***b* and **820***c* at the both sides of each slit **820** in a direction from the first casing **21** toward the second casing **22** (in a direction penetrating the surface of the flexible board **810**). Accordingly, this configuration can prevent application of high stress between the motor **3**, the sensor **7**, and the connection terminals **4**, even if a bonding member such as the solder **78** joins the flexible board **810** with the motor **3**, the sensor and connection terminals **4**, and it can prevent cracks or damages from occurring at the joint portions and can prevent the disconnection from occurring.
(99) Described next is a flexible board **811** included in a rotating device **1** according to still another embodiment. The rotating device **1** illustrated in FIG. **11**C differs from the rotating device **1** described above or the rotating device **1** illustrated in FIG. **11**A in that the housing **2** includes a different shape and the flexible board **811** includes a different shape. The flexible board **811** also includes a plurality of slits **830** as an example of a slit part. The slits **830** include two end portions **830***b* and **830***c* opposing each other in a direction crossing the slits **830** (e.g., in a direction in parallel with or orthogonal to the extension direction of the wires **833**). slits **830** are disposed between the two end portions **830***b* and **830***c*. Accordingly, the flexible board **810** includes discontinuous surfaces or discontinuous portions including the slits **830** in a direction crossing the slits **830**.
(100) The wires **833** at the flexible board **811** according the embodiment illustrated in FIG. **11**C includes bent portions in the middle of the wires **833**. Some of the slits **830** extend in parallel with a portion of the bent wires **833** extending in an axial direction of the connection terminals **4**. The other slits **830** extend in parallel with a portion of the bent wires **833** extending in a direction orthogonal to the axial direction of the connection terminals **4**.
(101) As illustrated in FIG. **11**C, the slits **830** are simple linear slits with no circular holes **820***a* (FIG. **11**B), for example. In other words, the shape of the slits provided to the flexible board **810**, **811** is illustrative and not limiting.
(102) As described above, the rotating device **1** includes the motor **3**, the board **700** provided with wires, and the flexible board **811** electrically connecting the board **700** provided with wires and the motor **3**. The flexible board **811** can include a terminal-side land **832** connected with the connection terminal **4**, a first land **831**, a wire **833** connecting the first land **831** with the terminal-side land **832**, and a slit part.
(103) A part of the flexible board **811** is slit, and the slit part may extend from the terminal-side

land **832** toward the first land **831**.
(104) The flexible board **811** may include a plurality of first lands including the first land **831**, a plurality of terminal-side lands including the terminal-side land **832**, and a plurality of wires **833** connecting the plurality of first lands with the plurality of terminal-side lands. The slit part may be disposed between the wires **833**.
(105) The terminal-side land **832** on the flexible board **811** may be connected with the connection terminal **4** or with a land on a board mounted with an electronic component, and the first land **831** on the flexible board **811** may be connected to a terminal of the sensor **7** or the motor terminal **33**.
(106) The rotating device **1** including the structure above can prevent breakage of the wire **833** or disconnection at the lands (first land **831** and terminal-side land **832**) if a stress is applied to the flexible board **811** due to the vibration of the rotating device **1** or an application of an external force.
(107) The slit part is not limited to the slit part extending from the terminal-side land **832** toward the first land **831**. The slit part may extend in a direction crossing the direction from the terminal-side land **832** toward the first land **831**. The slit part may be formed along the wires **833**.
(108) FIG. **12** is a perspective view of a connection terminal according to yet another embodiment. This connection terminal may be, for example, what is called a drawn-wire flanged-header connection terminal **400**. As illustrated in FIG. **12**, the connection terminal **400** includes a square-bar-shaped terminal body **410**, a bent portion **420** formed on an end portion side of the terminal body **410**, and a flange portion **430** in a middle of the connection terminal **400**. A plurality of connection terminals **400** can be collectively held by the holding member **9** (see FIG. **8**) with the flange portion **430** being engaged with the through-holes **91** of the holding member **9**. The drawn-wire flanged-header connection terminal is not limited to the bent-type terminal including the bent portion **420** as illustrated in FIG. **12**, and may be a linear type terminal with no bent portion **420**.
(109) The first corner **401** and the second corner **402** of the connection terminals **4** illustrated in FIG. **5** may be integrally formed with the housing **2**. This configuration can achieve a simple-shaped connection terminal **4** and can reduce the size of the rotating device **1**. Moreover, this configuration can easily and reliably increase the resistance of the connection terminals **4** to an external force exerted in the removal direction and can increase reliability.
(110) The embodiment described above provides the rotating device **1** described below. (1) The rotating device **1** includes a motor **3**, a connection terminal **4** electrically connecting the motor **3** with an external device, and a housing **2** accommodating the motor **3** and the connection terminal **4**. The connection terminal **4** includes a bent portion **40** bent in a direction crossing an insertion direction or a removal direction of the connection terminal **4**. The housing **2** includes a contact surface **216** in contact with the bent portion **40**.
(111) The rotating device **1** above can receive the force exerted in the removal direction of the connection terminal **4** at a surface of the housing **2**, and can easily and reliably increase the resistance of the connection terminal **4** to an external force exerted in the removal direction. (2) In the rotating device **1** in (1) above, the housing **2** includes a fixing part **215** configured to fix the bent portion **40**, and the fixing part **215** includes the contact surface **216**.
(112) The rotating device **1** described above can firmly fix the connection terminal **4** to the housing **2**. (3) In the rotating device **1** in (2) above, the fixing part **215** includes a recessed portion **217** engaged with the bent portion **40**.
(113) In the rotating device **1** described above, the bent portion **40** of the connection terminal **4** can be easily engaged with the recessed portion **217** by placing the connection terminal **4** at the fixing part **215**, and the connection terminal **4** can be easily fixed. (4) In the rotating device **1** in any one of (1) to (3) above, the connection terminal **4** is made of a bar-shaped member.
(114) The connection terminal **4** of the rotating device **1** described above can be easily bent and can be manufactured at lower material and manufacturing costs. Accordingly, the rotating device **1** can be manufactured at a lower cost. (5) In the rotating device **1** in any one of (1) to (4) above, the

connection terminal **4** includes a polygonal cross-section.
(115) The connection terminal **4** of the rotating device **1** above can easily include a bent portion, and can include a higher strength. (6) The rotating device **1** in any one of (1) to (5) above includes gears (transmission gears **6** and output gear **5**) configured to transmit the rotation of the motor **3** to an external device and the sensor **7** configured to detect the rotational angle of the output gear **5**, and a board (for example, a flexible board **8**) electrically connecting the motor **3**, the sensor **7**, and the connection terminal **4**. And the board is made of a flexible film.
(116) The rotating device **1** above can control rotational power transmitted to an external device, and if vibrations are applied to the board, flexibility of the board can absorb vibrations. If the board is electrically connected with the motor **3**, the sensor **7**, and the connection terminals **4** joined by solder or the like, application of high stress to the joint portions between these components can be avoided, and thus can prevent cracks or damages on the joint portions and can prevent the disconnection. (7) The rotating device **1** in any one of (1) to (6) above includes a plurality of terminals including the connection terminal **4**, and a holding member **9** including a plate shape and configured to hold the plurality of connection terminals **4** in a line.
(117) In the rotating device **1** above, the resistance of the connection terminals **4** to an external force exerted in the removal direction can be collectively increased. (8) In the rotating device **1** in any one of (1) to (7) above, the bent portion **40** includes a first corner **401** and a second corner **402** bent in opposite directions each other of a direction orthogonal to the removal direction, and includes a middle portion **403** connecting the first corner **401** with the second corner **402**. The connection terminal **4** of the rotating device **1** includes the bent portion **40**, a first extending portion **41** extending in the removal direction with an end extending toward outside of the housing **2** from the first corner **401**, and a second extending portion **42** extending in the removal direction from the second corner **402** toward the inside of the housing **2**.
(118) The connection terminal **4** of the rotating device **1** above can increase resistance to an external force exerted in the removal direction and can be easily manufactured by, for example, bending. Accordingly, the rotating device **1** can be manufactured at a lower cost. (9) The rotating device **1** includes a motor **3**, a plurality of connection terminals **4** electrically connecting the motor **3** with an external device, and a housing **2** accommodating the motor **3** and the connection terminals **4**. The holding member **9** configured to collectively hold a plurality of connection terminals **4** is disposed at the fixing part **215** of the housing **2**.
(119) The rotating device **1** above can collectively increase the resistance of the connection terminals **4** to an external force exerted in the removal direction regardless of the shape of the connection terminals **4**. (10) The rotating device **1** includes a motor **3**, a connection terminal **4** electrically connecting the motor **3** with an external device, and a housing **2** accommodating the motor **3** and the connection terminal **4**. The connection terminal **4** includes a first end portion electrically connected with the external device and a second end portion electrically connected with the motor **3**, and the first corner **401** and the second corner **402** disposed between a first end portion and the second end portion. The first corner **401** and the second corner **402** are in contact with the housing **2**.
(120) The rotating device **1** above can receive the force exerted in the removal direction of the connection terminal **4** by a surface of the housing **2**, and can easily and reliably increase the resistance of the connection terminal **4** to an external force exerted in the removal direction. (11) In the rotating device **1** in (10) above, the housing **2** includes a first housing **21** and a second housing **22**, and the first casing **21** includes an engagement portion **21*f*** and the second casing **22** includes an engagement portion **22*f***, and the first corner **401** and the second corner **402** are engaged with the engagement portion **21*f*** and the engagement portion **22*f*** of the housing **2**, respectively.
(121) The rotating device **1** above can increase the effect of the rotating device **1** in (10) above. (12) In the rotating device **1** in (11) above, the engagement portion **21*f*** and the engagement portion **22*f*** are configured by a combination of a recessed portion **22*e***, a protruding portion **22*c*** (**215*a***) and

a step portion **22***d*.
(122) The rotating device **1** above can implement a structure for increasing the resistance of the connection terminal to an external force exerted in the removal direction upon assembly of the housing. (13) The rotating device **1** in any one of (10) to (12) above includes a flexible board **8**, and the flexible board **8** electrically connects the connection terminal **4** with the motor **3**.
(123) The flexible board **8** of the rotating device **1** above can absorb vibrations, and this configuration can prevent application of a high stress to a joint portion of the connection terminal **4**, and thus can prevent cracks or damages at the joint portions and can prevent the disconnection in advance. (14) The rotating device **1** in (13) above includes a transmission gear **6**, an output gear **5**, and an angle sensor **7** configured to detect a rotational angle of the output gear **5**, and the flexible board **8** electrically connects the connection terminal **4**, the motor **3**, and the angle sensor **7**.
(124) The flexible board **8** of the rotating device **1** above can absorb vibrations, and this configuration can prevent application of a high stress to a joint portion of the connection terminal **4** with the motor **3** or the angle sensor **7**, and thus can prevent cracks or damages at the joint portions and can prevent the disconnection. (15) The rotating device **1** in any one of (10) to (14) above includes a plurality of connection terminals **4** including the connection terminal **4** and the holding member **9** configured to hold the plurality of connection terminals **4** lined in a predetermined direction.
(125) The rotating device **1** above can collectively and neatly hold the connection terminals **4** and can increase the resistance of the connection terminals **4** to an external force exerted in the removal direction. (16) In the rotating device **1**, includes a motor **3**, a connection terminal **4** electrically connecting the motor **3** with an external device, and a housing **2** accommodating the motor **3** and the connection terminal **4**. The connection terminal **4** includes a first end portion electrically connected with the external device, a second end portion electrically connected with the motor **3**, the first corner **401** as a first bent portion and the second corner **402** as a second bent portion disposed between the first end portion and the second end portion, and the first corner **401** and the second corner **402** are integrally formed with the housing **2**.
(126) The rotating device **1** above can achieve a simple-shaped connection terminal and can reduce the size of the rotating device. Moreover, this configuration can easily and reliably increase the resistance of the connection terminals **4** to an external force exerted in the removal direction and can increase reliability. (17) The rotating device **1** includes the motor **3**, the board **700** provided with a wire, and the flexible board **8** electrically connected with the motor **3** and the board **700** provided with the wire, and the flexible board **8** and the board **700** provided with the wire are overlapped via a solder **78** (bonding member) and bonded, and a first land (land portion) **803** included in the flexible board **8** including a hole **805**, and the solder **78** is in contact with a second land (land portion) **703** in the board **700** provided with the wire via the hole **805**.
(127) The flexible board **8** of the rotating device **1** above can be firmly bonded to the board **700** provided with the wire without lift with a simple structure, and this configuration can prevent, for example, disconnection of wires. (18) In the rotating device **1** in (17) above, the solder **78** (bonding member) is disposed and straggled between a portion of the first land **803** to a portion of the second land **703**.
(128) This configuration facilitates bonding using the solder **78** and can easily guide the solder **78** to the hole **805** of the first land **803**. (19) In the rotating device **1** in (17) or (18) above, the flexible board **8** includes a surface opposing the board **700** provided with the wire and a side portion, and the side portion includes a recess.
(129) This configuration allows the flexible board **8** to be more firmly bonded with the board **700** provided with the wire. (20) The rotating device **1** in any one of (17) to (19) above includes the connection terminal **4** electrically connecting the motor **3** with an external device, and the first land **803** of the flexible board **8** is electrically connected with the connection terminal **4**.
(130) This configuration can facilitate electrical connection between the motor **3** and the external

device via the flexible board **8** and the connection terminal **4**. Furthermore, vibrations are can be absorbed at the flexible board **8** when vibrations are generated, and, for example, disconnection of wires due to the vibrations can be prevented in advance. (21) In the rotating device **1** in (20) above, the flexible board **8** includes a bent portion **800** between an end portion connected with the connection terminal **4** and another end portion provided with the first land **803**.

(131) This configuration can absorb misalignment in the removal direction of the connection terminal **4** in connecting the flexible board **8** with the board **700** provided with the wire. (22) In the rotating device **1** in (20) or (21) above, a part of the flexible board **810**, **811** is slit, and the slit part extends from the terminal-side land **832** connected with the connection terminal **4** toward the first land **831**.

(132) This configuration can prevent breakage of the wire **833** or disconnection at the lands (first land **831** and terminal-side land **832**) if a stress is applied to the flexible board **810**, **811** due to the vibration of the rotating device **1** or an application of an external force. (23) In the rotating device **1** in (22) above, the flexible board **810**, **811** includes a plurality of first lands including the first land **831**, a plurality of terminal-side lands including the terminal-side land **832**, and a plurality of wires **833** connecting the plurality of first lands with the plurality of terminal-side lands, and the slit part is disposed between the plurality of wires **833**. (24) The rotating device **1** in (22) or (23) above further includes gears (transmission gears **6** and output gear **5**) configured to transmit the rotation of the motor **3** to an external device and the sensor **7** configured to detect the rotational angle or the number of rotations of the output gear **5**. The terminal-side land **832** of the flexible board **810**, **811** is connected with the connection terminal **4** or with a land of a board provided with an electronic component and the first land **831** of the flexible board **810**, **811** is connected with a terminal of the sensor **7** or with a motor terminal **33**. (25) In the rotating device **1** in (24) above, the flexible board **8** electrically connects the motor **3**, the sensor **7**, and the connection terminal **4**.

(133) This configuration can absorb vibrations, if vibrations of the motor **3** are transmitted to the flexible board **8**. Accordingly, application of high stress to the joint portions between the motor **3**, the sensor **7**, or the connection terminal **4** and the flexible board **8** can be prevented, and thus and can prevent disconnection due to cracks or damages on the joint portions. (26) In the rotating device **1** in (25) above, the sensor **7** includes a brush **75** formed with conductive member configured to be rotated together with the output gear **5**, and the board **700** provided with the wire. The board **700** provided with the wire includes a plurality of wires (first wire pattern **701** and second wire pattern **702**) in electrical contact with the brush **75**.

(134) This configuration makes the relation between a displacement in contact positions of the brush **75** relative to the wires and a change in voltage linearly proportional. The rotational angle of the output gear **5** can be easily detected from the change in voltage. (27) The rotating device **1** includes the output gear **5** configured to transmit rotational power to an external device, the motor **3** configured to rotate the output gear **5**, the sensor **7** configured to detect the number of rotations or the rotational angle of the output gear **5**, a gear and the housing **2** accommodating the gear, the motor **3**, and the sensor **7**. The sensor **7** includes the brush **75** configured to be rotated together with the output gear **5**, and the board **700** provided with an arc-shaped first wire pattern **701** and the arc-shaped second wire pattern **702** in electrical contact with the brush **75**. The board **700** provided with the wire is provided with a first hole **710** disposed at the center of the arc-shaped wire patterns and providing a interval fit to a rotating shaft **51** of the output gear **5**, and a second hole **720**, a protruding portion **218** provided to the housing **2** passing through the second hole **720**.

(135) This configuration allows for accurate positioning and fixing of the board **700** provided with the wire with a difference between the center of the arc-shaped wire patterns and the center of rotation of the brush **75** as small as possible. Accordingly, the linearity (linear proportional relation) between the displacement in contact positions of the brush **75** relative to the wire patterns and the change in voltage is not degraded. (28) In the rotating device **1** in (27) above, the housing **2** includes the first casing **21** and the second casing **22** opposing each other. The first casing **21**

accommodates the output gear **5**, the motor **3**, and the sensor **7**. The second casing **22** includes a first step portion **201** disposed between the first casing **21** and the second casing **22**. The first step portion **201** supports the board **700** provided with the wire.
(136) This structure allows the board **700** provided with wires to be disposed more accurately parallel to the housing **2**. (29) In the rotating device **1** in (28) above, the rotating shaft **51** of the output gear **5** includes a step portion **515**, the board **700** provided with the wire being supported between the second casing **22** and the step portion **515**.
(137) In this structure, the board **700** provided with wires is supported by the step portion **515** by inserting the rotating shaft **51** in the first hole **710**. This configuration facilitates positioning and fixing of the board **700** provided with wires. (30) In the rotating device **1** in (29) above, the second casing **22** includes a tubular support portion **228** configured to rotatably support an end portion of the rotating shaft **51**, a second step portion **202** being formed at the tubular support portion **228**. The first hole **710** of the board **700** provided with the wire is disposed between the second step portion **202** and the step portion **515** provided at the rotating shaft **51** of the output gear **5**.
(138) This configuration allows for more accurate positioning and fixing of the board **700** provided with wires. (31) In the rotating device **1** in (30) above, the protruding portion **218** is provided to the first casing **21**, and the board **700** provided with the wire is supported by the protruding portion **218**, the first step portion **201**, and the second step portion **202**.
(139) This configuration allows the board **700** provided with wires to be disposed more accurately parallel to the housing **2**. (32) In the rotating device **1** in (30) or (31) above, a interval within certain tolerances is provided between the rotating shaft **51** of the output gear **5** and the first hole **710**, between the protruding portion **218** provided to the housing **2** and the second hole **720**, and between the board **700** provided with the wire and the protruding portion **201***a*, the first step portion **201**, and the second step portion **202**.
(140) This configuration can prevent degradation of linearity (linear proportional relation) between a displacement in contact positions of the brush **75** relative to the wire patterns and a change in voltage as much as possible, and thus can control the rotational operation more accurately.
(141) AT the embodiments as described above the crank-shaped connection terminal **4** is used as a terminal. However, for example, the holding member **9** in (9) above can hold terminals in any shape. The bonding structure between the board **700** provided with wires and the flexible board **8** using the solder **78** such as (10) above does not at all limit the shape of the terminal.
(142) This application claims the benefit of Japanese Patent Application No. 2018-102696 filed May 29, 2018, Japanese Patent Application No. 2018-102697 filed May 29, 2018, and Japanese Patent Application No. 2018-102698 filed May 29, 2018.
REFERENCE SIGNS LIST
(143) **1** Rotating device **2** Housing **3** Motor **4** Connection terminal **5** Output gear **6** Transmission gear **7** Sensor **8** Flexible board **9** Holding member **20** Connector portion **21** First casing **22** Second casing **23**, **24**, **25**, **26** Mounting portion **30** Body **31** Rotating shaft **32** Worm gear **33** Motor terminal **40** Bent portion **41** First extending portion **42** Second extending portion **51** Rotating shaft **52** Gear body **61** First transmission gear **62** Second transmission gear **75** Brush **78** Solder (bonding member) **81** First planar portion **82** Second planar portion **83** Third planar portion **91** Through-hole **201** First step portion **202** Second step portion **210** First surface portion **211** First sidewall **212** Opening **215** Fixing part **215***a* First fixing part **215***b* Second fixing part **216** Contact surface **216***a* First contact surface **216***b* Second contact surface **217** Recessed portion **218** Protruding portion **219** Projecting portion **219***a* Wall portion **220** Second surface portion **221** Second sidewall **222** Opening **223** Engagement portion **224** Engagement step portion (step portion) **228** Support **229** Projecting portion **401** First corner **402** Second corner **403** Middle portion **510** Output part **511** Teeth **512** End portion **513** Teeth **515** Step portion **611** First large diameter portion **612** First small diameter portion **621** Second large diameter portion **622** Second small diameter portion **700** Board provided with wires **701** First wire pattern (wire) **702** Second wire pattern (wire) **703** Second land

703*a*, **703***b*, **703***c* Land portion **710** First hole **720** Second hole **730** Third hole **751**, **752** Contact **800** Bent portion **803** First land **805** Hole **810**, **811** Flexible board **831** First land **832** Terminal-side land **833** Wire a**1** First support a**2** Second support a**3** Third support

Claims

1. A rotary apparatus comprising: a sensor including a terminal; a motor including a motor terminal; a gear; a film including a conductor; and a first connection terminal electrically connecting the terminal of the sensor via the film, a second connection terminal electrically connecting the motor terminal via the film, wherein the film includes a first portion electrically connecting the terminal of the sensor and the first connection terminal, a second portion electrically connecting the motor terminal and a third portion connecting the first portion and the second portion, the third portion electrically connects with the second connection terminal, a bent portion of the first portion includes one slit or a plurality of slits extending from the terminal of the sensor toward the first connection terminal, and the conductor is arranged between the plurality of the slits.
2. A vehicle comprising: an air conditioning system including: a rotary apparatus according to claim 1; and a louver controlled by the rotary apparatus.